10. Develop a JAVA program to create a package named mypack and import & implement it in a suitable class

Step 1: Create a Package

Create a directory named mypack (or any name you prefer) on your file system. Inside the mypack directory, create a Java file named MyClass.java.

Step 2: Define the Class in the Package. Open MyClass.java and add the following code:

```
package mypack;
public class MyClass
{
public void
display ()
{
System.out.println("This is a method from the mypack
package.");
}
}
```

Step 3: Create a Main Program

Create a new Java file (outside the mypack directory) named MainProgram.java and add the following code:

```
import mypack.MyClass;
public class MainProgram
{
public static void main (String[] args)
{
MyClass obj = new MyClass ();
obj.display ();
}
}
```

Step 4: Compile and Run

Open a terminal or command prompt and navigate to the directory containing both the mypack directory and MainProgram.java.

Compile both files using the following command:

javacmypack/MyClass.java MainProgram.java

Run the program with the command:

java MainProgram

Output:

This is a method from the mypack package.

11.Write a program to illustrate creation of threads using runnable class. (start method start each of the newly created thread. Inside the run method there is sleep() for suspend the thread for 500 milliseconds)

Save Filename as: ThreadMain.java

Solution:-

```
class MyRunnable implements Runnable
{
@Override
public void run()
{
Try
{
System.out.println("Thread " +
Thread.currentThread().getId() + " is running.");
Thread.sleep(500);
}
catch (InterruptedException e)
{
System.out.println("Thread interrupted: " +
e.getMessage());
}
}
}
public class ThreadMain
{
public static void main(String[] args)
{
int numThreads = 5;
for (int i = 0; i<numThreads; i++)
{
Thread thread = new Thread(new MyRunnable());
```

```
thread.start();
}
}
}
```

Compile As: javacThreadMain.java

Run As: java ThreadMain

Output:

Thread 10 is running.

Thread 12 is running.

Thread 14 is running.

Thread 13 is running.

Thread 11 is running.

12. Develop a program to create a class MyThread in this class a constructor, call the base class constructor, using super and start the thread. The run method of the class starts after this. It can be observed that both main thread and created child thread are executed concurrently.

Save Filename as: ThreadConstructor.java

Solution:-

```
class MyThread extends Thread
{
MyThread (String name)
{
super (name); // Call the base class constructor with thread
name start (); // Start the thread
}
public void
run ()
{
try
{
for (int i = 5; i > 0; i--)
{
System.out.println (getName () + ": " + i);
Thread.sleep (1000);
}
}
catch (InterruptedException e)
{
System.out.println (getName () + " interrupted.");
}
System.out.println (getName () + " exiting.");
}
}
public class ThreadConstructor
```

```
{
public static void main (String[] args)
{
new MyThread ("Child Thread");
try
{
for (int i = 5; i > 0; i--)
{
System.out.println ("Main Thread: " + i);
Thread.sleep (2000);
}
}
catch (InterruptedException e)
{
System.out.println ("Main Thread interrupted.");
}
System.out.println ("Main Thread exiting.");
}
}
```

Compile As: javacThreadConstructor.java

Run As: java ThreadConstructor

Output:

Main Thread: 5

Child Thread: 5

Child Thread: 4

Main Thread: 4

Child Thread: 3

Child Thread: 2

Main Thread: 3

Child Thread: 1

Child Thread exiting.

Main Thread: 2

Main Thread: 1

Main Thread exiting